



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(MECHANICAL ENGINEERING)

EMM 314: MANUFACTURING WORK STUDY

1ST SEMESTER 2022/2023

TIME: 2 HOURS

Exam By Dr. Wanjau wa Kabecha.

INSTRUCTIONS: Answer question One and any other TWO questions.

1. (a) State briefly the reasons why we apply methods study in manufacturing or in services. (8 Marks)
- (b) Give an outline of what can be achieved through work measurement. (9 Marks)
- (c) An assembly line provides the desired output rate for an eight hour shift. Calculate the following:
- (i) Total work content
 - (ii) Maximum shift output
 - (iii) Number of work stations
 - (iv) Efficiency
 - (v) Idleness of the line.

Task	1	2	3	4	5	6
	A	B C	D	E F	G H	I
Time	4	1.5 1.5	3.5	1.5 2.5	2 1.5	4
(Min)						

(10 Marks)

- (d) What factors do we consider in selecting job for direct time study? (3 marks)
2. (a) A work sampling study was conducted to establish the standard time for an operation. The observations of the study were as follows:
- No of observation 160
 - Manual work 14
 - Machine work 106
 - Machine idle time 40
 - Performance raking 80%
 - No of parts produced 36
 - Allowance for personal needs and fatigue 10%
- The study was conducted for three days with 8 hours per day.

Calculate the standard time per piece.

(8 Marks)

(b) What are the advantages of work sampling as compared to time study? (9 Marks)

(c) Why should work study be considered an aid in management? (3 Marks)

3 (a) State briefly the reasons why a job should be selected for methods study. (10 Marks)

(b) Discuss briefly the areas we should focus on in improving work methods. (5 Marks)

(c) Give reasons for excess work content with respect to defects in design or specifications of a product. (5 Marks)

4. (a) A machine operation begins when the component is brought from the stores 10 metres away. The component is loaded of the machine, which takes two minutes. The component is machined, which takes five minutes. The component is then moved for inspection to an inspections bench, twelve metres away. It has to wait for fifteen minutes for the inspector to be free from the previous job. The component is checked for accuracy, which takes two minutes. The component is taken back to the stores twelve metres away from the inspection bench and stored in a rack.

Construct a process chart for this operation. Analyse this operation and using the principles of motion economy, suggest improvements that could make this operation simpler. (10 marks)

(b) Give an outline of the areas where methods study would have an impact in improving work methods. (6 marks)

(c) Explain the predetermined time study approach to work measurement. (4 marks)

5. (a) Give an outline of the principles of motion economy as they relate to the use of human arms. (4 marks)

(b) The observed times and performance rating for five elements are as follows:

Element	1	2	3	4	5
Rating	80	100	90	80	110
Frequency	1	1	1	1	1
Time	0.25	0.09	2.80	0.05	0.15

Calculate the standard time if the rest and personal allowance is 12%. (4 marks)

(c) The following table shows a time study data. The times shown are continuous watch readings in minutes. Initial setting of stop watch is at 0.00

Element	Cycle Time			Performance rating
	1	2	3	
Get two cases	0.5	4.2	8.6	1.05
Put parts into cases	1.5	5.7	9.9	1.15
Clamp two parts in position	3.8	8.1	12.6	0.95

From Industrial Engineering & Production Management, By Martand T Telsang, Pg 118

Take relaxation allowance as 15% and find standard time.

(4 times)

(d) Which factors are likely to cause variations in times for an element during time study?

(6 marks)

(e) What are the precautions that should be taken during stop watch timing study?

(2 marks)

Element	Cycle time			Avg. time	Normal time
	1	2	3		
Get two cases	0.5	0.4	0.5	0.466	0.49
Put parts into cases	1.0	1.5	1.3	1.266	1.456
Clamp two parts in position	2.3	2.4	2.7	2.466	2.343

$$\begin{aligned}
 \text{Std time} &= \text{Normal time} (1 + 0.15) \\
 &= (0.49 + 1.456 + 2.343) (1.15) \\
 &= 4.932 \text{ min.}
 \end{aligned}$$